



برمجة الشبكات

12:00-2:00

13/6/2023

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Faculty of Computers & Information, Assiut University

Course code: IT432

4th Level

Final Exam

Duration: 2 hours

Total Marks: 50

Reviewer: (Add the name)

This exam for the following program(s) : (Add the program's name(s))

* Required

* This form will record your name, please fill your name.

1

* الإسم الرباعي (بالعربي فقط)

2

* رقم الجلوس

3

* المستوى

- ☐ الاول
- ☐ الثاني
- ☐ الثالث
- ☐ الرابع 2013
- ☐ الرابع 2014
- ☐ الرابع 2015
- ☐ الرابع 2016
- ☐ الرابع 2017
- ☐ الرابع 2018
- ☐ الرابع 2019

4

* البرنامج

- ☐ عام
- ☐ بايو
- ☐ هندسة
- ☐ ذكاء صناعي

5

* رقم المعمل

- ☐ ج٠
- ☐ د٠
- ☐ ه٠
- ☐ اب
- ☐ اج
- ☐ اد
- ☐ اه
- ☐ أ٢
- ☐ ب٢
- ☐ ج٢
- ☐ د٢
- ☐ ه٢
- ☐ أ٣
- ☐ ب٣
- ☐ ج٣
- ☐ د٣
- ☐ ه٣

- ☐ أ
- ☐ ب
- ☐ ج
- ☐ د
- ☐ هـ

6

* رقم الكمبيوتر

7

* الكود (قد تمت مراجعة بيانات الطالب ورقم الجلوس)

8

The following are network applications Except (1 Point)

- ☐ Text, Audio and video chat.
- ☐ Video broadcast and multicast.
- ☐ Web browser
- ☐ Compiler

9

All of these are Network benefits Except (1 Point)

- ☐ Sharing resources
- ☐ Load Sharing/utilization
- ☐ High reliability
- ☐ Enhance Security

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Circuit switched networks are ... (1 Point)

- ☐ dedicated circuit per call
- ☐ performance (guaranteed)
- ☐ resources used as needed
- ☐ setup required

11

Packet-switched networks are ... (1 Point)

- ☐ dedicated circuit per call
- ☐ performance (guaranteed)
- ☐ resources used as needed
- ☐ setup required

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Common internet applications (1 Point)

- ☐ Client/server
- ☐ Email
- ☐ P2P architecture
- ☐ Resource distribution (DNS)

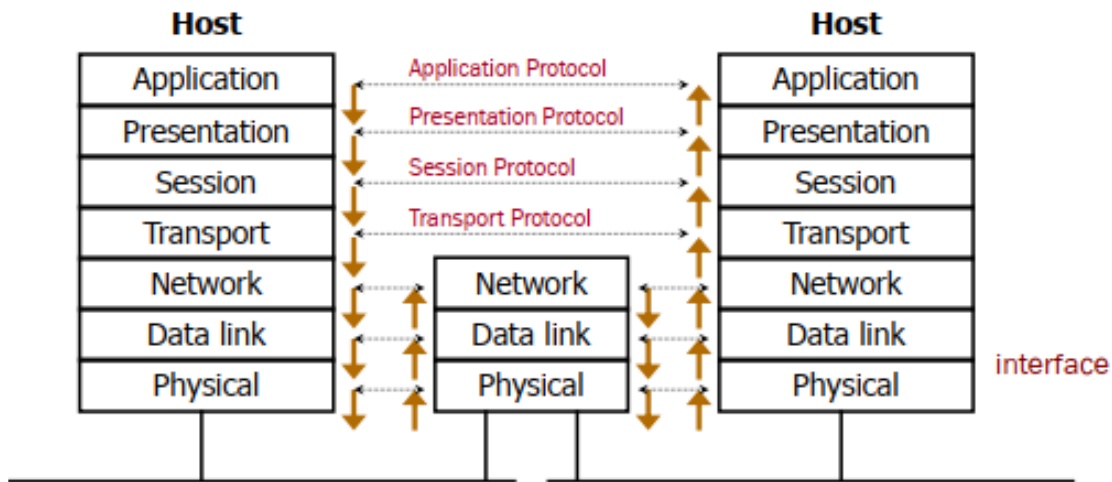
13

Sensor Networks (1 Point)

- ☐ Small devices equipped by sensor and connected to the
- ☐ May include an environmental actuation
- ☐ Run on batteries
- ☐ Location don't matter

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The host with full layers may be (choose all the correct answers) (2 Points)



- ☐ Router
- ☐ Server
- ☐ Switch
- ☐ Client

15

Example of protocols operates on layer 5 in the OSI Model (1 Point)

☐ RPC☐ XNS☐ FTP☐ TCP

16

Major INTEROPERABILITY advantages (1 Point)

☐ Code Reuse☐ Eases maintenance, updating of system☐ Communication☐ Reliability

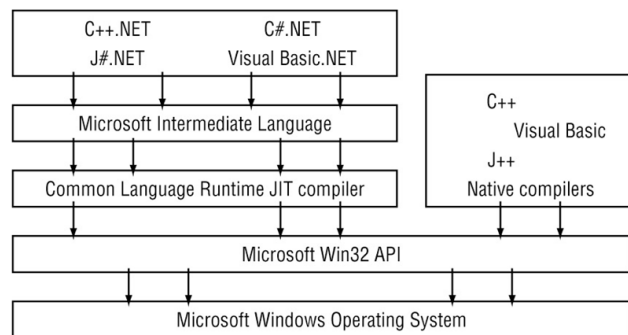
17

The most Basic network Architecture is in terms of setup and cost
(1 Point)

- ☐ Client/server
- ☐ P2P

18

The layers which is uniquely exist to mimic managed code is
(2 Points)



- ☐ MSIL
- ☐ CLR JIT complier
- ☐ win32 API
- ☐ Windows OS

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The most common managed programming language from Microsoft (1 Point)

- ☐ C#.net
- ☐ VB. net
- ☐ C++
- ☐ JAVA

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The upper stream is an _____ and the lower is _____ (2 Points)



- ☐ Output , input1
- ☐ input , output
- ☐ input, input
- ☐ output,output

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Reads primitive data types as binary values in a specific encoding (1 Point)

- ☐ BinaryReader
- ☐ BinaryWriter
- ☐ FileStream
- ☐ StreamReader

22

to use the "using " control structure the type must (1 Point)

- ☐ implement the interface IDisposable
- ☐ implement the interface ICompiler
- ☐ implement the interface IDisposed
- ☐ implement the interface Iformatter

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ADAPTER CLASSES are set of classes provided by FCL to support reading and writing in higher level such as (1 Point)

☐ StreamReader☐ BinaryReader☐ FileStream☐ StreamWriter

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Identify the output (1 Point)

```
using System;
using System.IO;

public class BinaryWriteSimpleTypes{

    public static void Main(){
        string fn = "simple-types.bin";

        using(BinaryWriter bw =
            new BinaryWriter( new FileStream(fn, FileMode.Create))){
            bw.Write(5); // 4 bytes
            bw.Write(5.5); // 8 bytes
            bw.Write(5555M); // 16 bytes (decimal dt)
            bw.Write(5==6); // 1 bytes
        }

        FileInfo fi = new FileInfo(fn);
        Console.WriteLine("Length of {0}: {1}", fn, fi.Length);
    }
}
```

☐ Length of simple-types.bin: 29☐ Length of simple-types.bin: 33☐ Length of simple-types.bin: 25☐ Length of simple-types.bin: 17

25

_____ is the best way to save objects in streams so that we can "send" it any where or even save it immediately (1 Point)

- ☐ Serialization
- ☐ Saving to file
- ☐ Inserting into database
- ☐ sending through networks

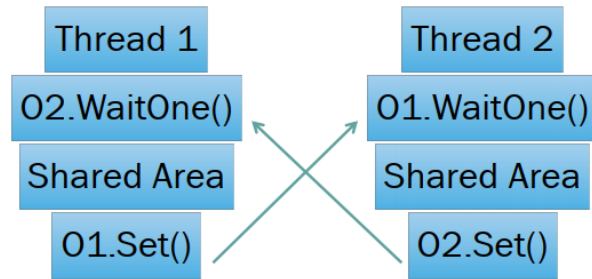
26

the [OnSerialized()] attribute is to identify the method invoked ... (1 Point)

- ☐ After obj Serialization
- ☐ After obj DeSerialization
- ☐ During obj Serialization
- ☐ During obj DeSerialization

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This pic shows how to use AutoResetEvent, why we are using that during managing threads ? (1 Point)



- ☐ so that we can avoid RACE CONDITION?
- ☐ to control shared variable between objects
- ☐ to lock the code
- ☐ none of the answers are correct

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Connection oriented Socket obj parameters are : (1 Point)

- ☐ AddressFamily.InterNetwork
- ☐ [SocketType.Stream](#)
- ☐ SocketType.Dgram
- ☐ ProtocolType.Tcp
- ☐ ProtocolType.Udp
- ☐ ProtocolType.Raw

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Example Socket Blocking Call (2 Points)

☐ RecvFrom()☐ Bind()☐ Sendto()☐ Accept()

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Identify the line number where the compression process is performed (2 Points)

```
1 private void btnBackUp_Click(object sender, EventArgs e) {  
2     if (dlgOpenFile.ShowDialog(this) == DialogResult.OK)  
3         GZipStream gz = new GZipStream(  
4             new FileStream(dlgOpenFile.FileName + ".QZip.Compressed",  
5                 CompressionMode.Compress);  
6         FileStream fs = new FileStream(dlgOpenFile.FileName,  
7             txtSizeBefore.Text = fs.Length.ToString();  
8             progressBar1.Maximum = (int)fs.Length;  
9             progressBar1.Value = (int)fs.Length;  
10            StreamWriter sw = new StreamWriter(gz);  
11            sw.Write(new StreamReader(fs).ReadToEnd());  
12            sw.Flush(); sw.Close();  
13            txtSizeAfter.Text = new FileStream(dlgOpenFile.FileName +  
14                progressBar2.Maximum = (int)fs.Length;  
15                progressBar2.Value = (int.Parse(txtSizeAfter.Text)  
16            }  
17        }  
18    }  
19 }
```

☐ line 10☐ line 11☐ line 12☐ line 13

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All of these options are correct regarding broadcast **Except** (2 Points)

- ☐ To send a single packet of information that can be accessible by every device on the network.
- ☐ Two types: local and global
- ☐ The main goal is to limit the broadcasting network to developers' network and other are unaffected.
- ☐ Because TCP communication requires that two devices have a dedicated connection, it is possible to send broadcast messages in a strictly TCP environment.
- ☐ Instead, UDP packets must be used because that protocol has the capability of sending messages without a specific connection being defined

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The least significant bits in an ip address represents (2 Points)

- ☐ Net ID
- ☐ Host ID
- ☐ Class ID
- ☐ none of the options are correct

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look and choose (2 Points)

```
1 using System;
2 using System.Net;
3 using System.Net.Sockets;
4 using System.Text;
5 class BadBroadcast
6 {
7     public static void Main()
8     {
9         Socket sock = new Socket(AddressFamily.InterNetwork, SocketType
10             ProtocolType.Udp);
11         IPEndPoint iep = new IPEndPoint(IPAddress.Broadcast, 9050);
12         byte[] data = Encoding.ASCII.GetBytes("This is a test message");
13         sock.SendTo(data, iep);
14         sock.Close();
15     }
16 }
17
```

- ☐ this program will broadcast
- ☐ this program will throw an exception

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We can receive the broad cast messages by ... (2 Points)

- ☐ using Wireshark
- ☐ using Ethereal
- ☐ coding a udp receiver
- ☐ using browser

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The Ip address and port number for common shared hardware in a network are .. (2 Points)

- ☐ Fixed
- ☐ preconfigured
- ☐ Advertised
- ☐ none of the options are correct

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helper classes used for socket programming (2 Points)

☐ TcpListener☐ UdpClient☐ TcpClient☐ GzipStream

37

Threads on simple PC run (2 Points)

☐ Concurrently☐ Sequentially☐ Simlitiniously☐ on parallel

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We may need threading during writing network programming to . . . (2 Points)

- ☐ Improve Application performance
- ☐ Enable our server to serve multi clients
- ☐ made our GUI don't freeze
- ☐ all the answers are correct

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select the steps that we may need to thread them in this high level view of a client (2 Points)

Create a socket

Setup the server address

Connect to the server

Read/write data

Shutdown connection

- ☐ Create a socket
- ☐ Setup the server address
- ☐ Connect to the server
- ☐ Read/write data
- ☐ Shutdown connection

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select the steps that we may need to thread them in this high level view of a multithreaded - responsive GUI server (2 Points)

Create a socket

Bind the socket

Listen for connections

Accept new client connections

Read/write to client connections

Shutdown connection

- ☐ Create a socket
- ☐ Bind the socket
- ☐ Listen for connections
- ☐ Accept new client connections
- ☐ Read/write to client connections
- ☐ Shutdown connection

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Common (Major)socket blocking calls . . . (2 Points)

- ☐ Accept
- ☐ Receive
- ☐ Send
- ☐ Send To

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